

Module

3

Operating System Fundamentals - Theoretical Content



Part 5: System Administration Basics



Domain Controllers - Windows

- Central Authentication Server:
 - Stores user accounts
 - Authenticates logins
 - Manages computers
 - Applies Group Policy
 - Critical infrastructure

Installing and Joining Domain

- Process:
 - Install Server with AD role
 - Promote to Domain Controller
 - Create domain (company.local)
 - Join workstations to domain
 - Users log in with domain credentials

Group Policy Objects (GPO)

- Configuration management:
 - Security policies
 - Software deployment
 - Scripts at login/startup
 - Registry modifications
 - Scales to thousands

Windows Server Roles

- Common roles:
 - AD Domain Services - Domain controller
 - DNS - Name resolution
 - DHCP - IP assignment
 - File Services - Network shares
 - IIS - Web server

Linux Server Administration

- Common tasks:
 - User/group management
 - Service configuration
 - Network setup
 - Package installation
 - Log monitoring
 - Security updates
- All via command line!

Linux File Sharing (Samba)

- Samba = Windows compatibility:
 - Linux server, Windows clients
 - Uses SMB/CIFS protocol
 - Shares files and printers
 - Can join a Windows domain

Backup Strategies

Three types:

- Full - Everything (slow, large)
- Incremental - Changes since last backup
- Differential - Changes since last full

Choose based on needs

Volume Shadow Copy (VSS)

- Windows backup technology
 - Point-in-time file snapshots
 - Restore previous versions
 - "Undo" accidental changes
 - Ransomware recovery

Disaster Recovery Planning

- When worst happens:
 - Recovery Time Objective (RTO)
 - Recovery Point Objective (RPO)
 - Documented procedures
 - Regular testing
 - Offsite backups

Part 6: Security Event Management



Windows Event Logs - Deep Dive

Three main logs:

- Application - Program events
- System - OS events
- Security - Authentication, auditing

Each entry has: ID, Level, Source, Message

Critical Windows Event IDs

Event ID	Meaning	Why Important
4624	Successful login	Track access
4625	Failed login	Detect brute
4672	Admin privileges used	Monitor elevation
4688	Process created	See what executed
4698	Scheduled task created	Persistence method
7045	Service installed	Malware technique

Configuring Windows Auditing

- What to audit:
 - Logon events (success and failure)
 - Privilege use
 - Process tracking
 - Object access
 - Policy changes
- Configure via Group Policy

Linux System Logs

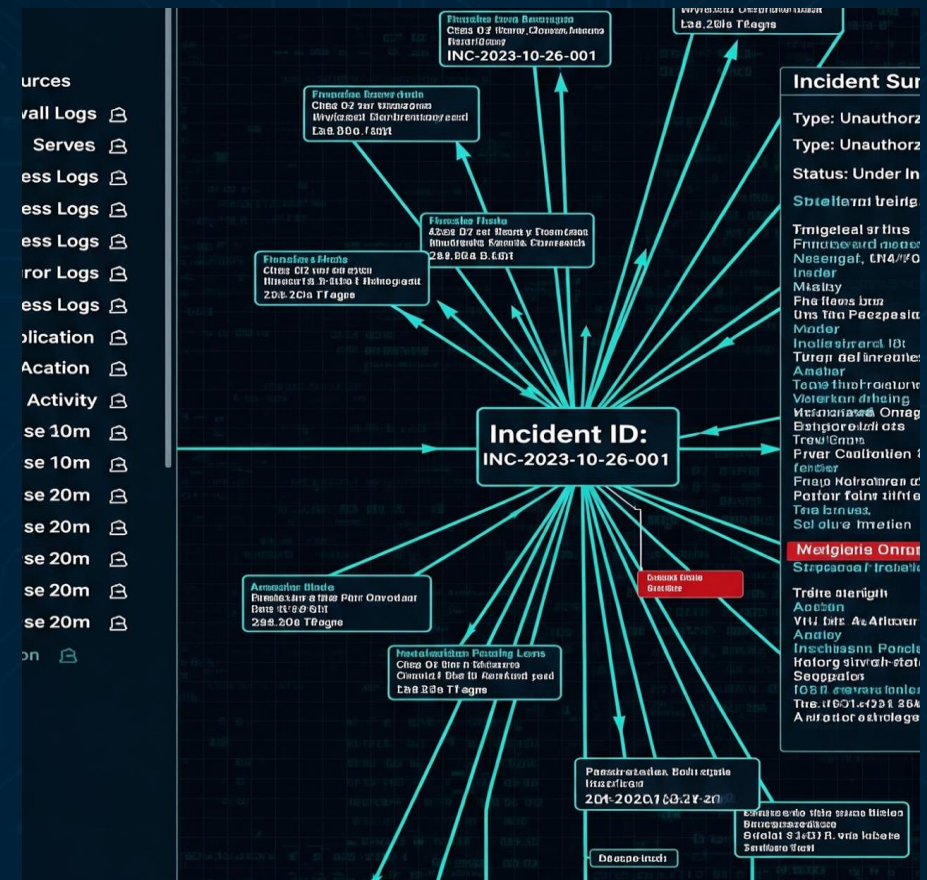
- Key log files:
 - `/var/log/syslog` - Everything
 - `/var/log/auth.log` - Logins, sudo
 - `/var/log/kern.log` - Kernel
 - `/var/log/apache2/` - Web server
 - `/var/log/mysql/` - Database

Reading Linux Logs

- Log entry format:
Oct 20 10:23:45 hostname process[PID]: message
 - Components: Date, Time, Host, Process, Message
- Tools for analysis: Use grep, awk, tail to analyze log entries.

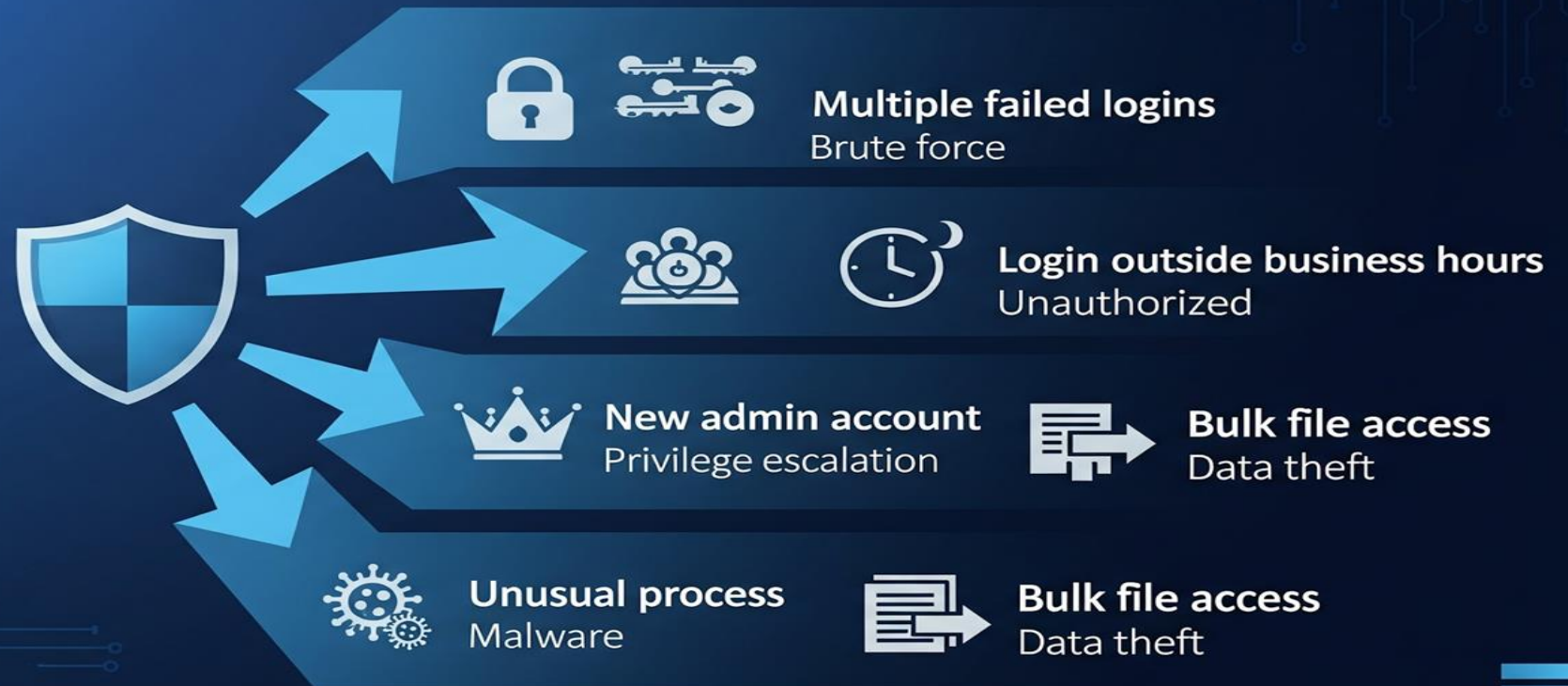
Security Event Correlation

- Connect the dots:
 - Same timestamp across logs
 - Same user/IP in multiple events
 - Sequence of related events
 - Build timeline of incident



Suspicious Event Patterns

Watch for:



Thank you!!